Welcome to Pydantic | Pydantic Docs

Welcome to Pydantic

Documentation for version: v2.12.5.

Pydantic is the most widely used data validation library for Python.

Fast and extensible, Pydantic plays nicely with your linters/IDE/brain. Define how data should be in pure, canonical Python 3.9+; validate it with Pydantic.

ⓘMonitor Pydantic with Pydantic Logfire 🔥

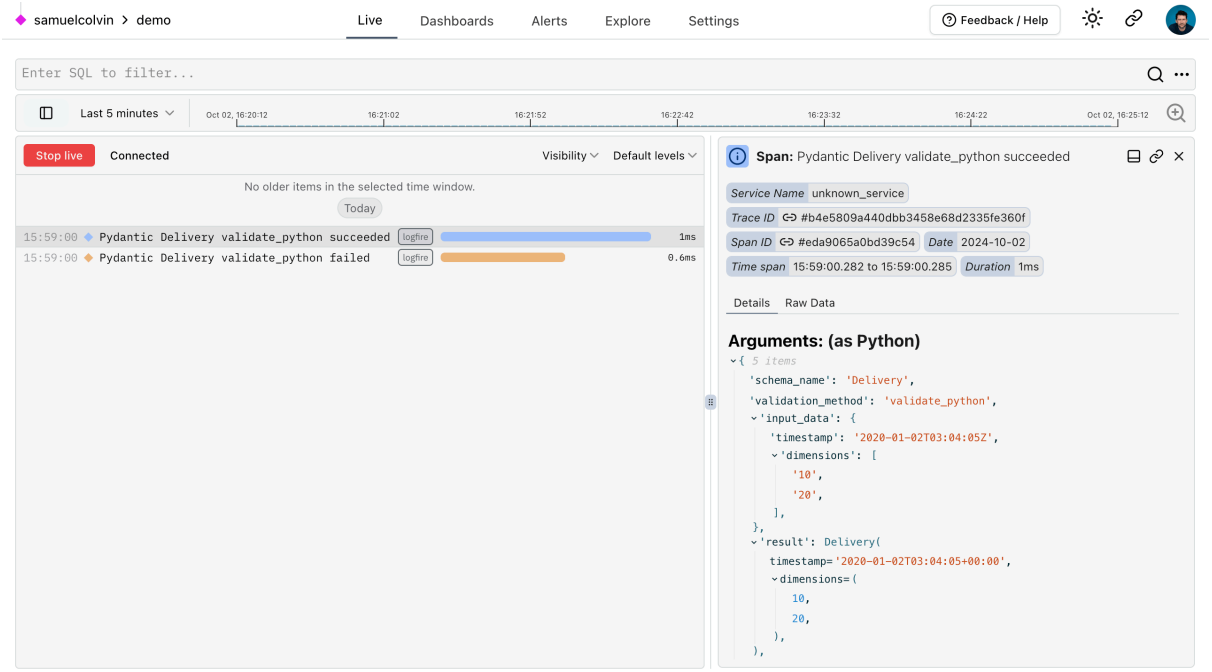
Pydantic Logfire is an application monitoring tool that is as simple to use and powerful as Pydantic itself.

Logfire integrates with many popular Python libraries including FastAPI, OpenAI and Pydantic itself, so you can use Logfire to monitor Pydantic validations and understand why some inputs fail validation:

Monitoring Pydantic with Logfire

```
from datetime import datetimeimport logfirefrom pydantic import
BaseModellogfire.configure()logfire.instrument_pydantic()  class
Delivery(BaseModel):  timestamp: datetime  dimensions: tuple[int, int]# 
this will record details of a successful validation to logfirem =
Delivery(timestamp='2020-01-02T03:04:05Z', dimensions=['10',
'20'])print(repr(m.timestamp))#> datetime.datetime(2020, 1, 2, 3, 4, 5,
tzinfo=TzInfo(UTC))print(m.dimensions)#> (10,
20)Delivery(timestamp='2020-01-02T03:04:05Z', dimensions=['10'])
```

Would give you a view like this in the Logfire platform:



This is just a toy example, but hopefully makes clear the potential value of instrumenting a more complex application.

Learn more about Pydantic Logfire

Sign up for our newsletter, *The Pydantic Stack*, with updates & tutorials on Pydantic, Logfire, and Pydantic AI:

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Why use Pydantic?

- **Powered by type hints** — with Pydantic, schema validation and serialization are controlled by type annotations; less to learn, less code to write, and integration with your IDE and static analysis tools. Learn more…
- **Speed** — Pydantic's core validation logic is written in Rust. As a result, Pydantic is among the fastest data validation libraries for Python. Learn more…
- **JSON Schema** — Pydantic models can emit JSON Schema, allowing for easy integration with other tools. Learn more…
- **Strict** and **Lax** mode — Pydantic can run in either strict mode (where data is not converted) or lax mode where Pydantic tries to coerce data to the correct type where appropriate. Learn more…
- **Dataclasses**, **TypedDicts** and more — Pydantic supports validation of many standard library types including `dataclass` and `TypedDict`. Learn more…
- **Customisation** — Pydantic allows custom validators and serializers to alter how data is processed in many powerful ways. Learn more…
- **Ecosystem** — around 8,000 packages on PyPI use Pydantic, including massively popular libraries like *FastAPI*, *huggingface*, *Django Ninja*, *SQLModel*, & *LangChain*. Learn more…
- **Battle tested** — Pydantic is downloaded over 360M times/month and is used by all FAANG companies and 20 of the 25 largest companies on NASDAQ. If you're trying to do something with Pydantic, someone else has probably already done it. Learn more…

Installing Pydantic is as simple as: `pip install pydantic`

Pydantic examples

To see Pydantic at work, let's start with a simple example, creating a custom class that inherits from `BaseModel`:

Validation Successful

```
from datetime import datetimefrom pydantic import BaseModel,
PositiveIntclass User(BaseModel):  id: int  name: str = 'John
Doe'  signup_ts: datetime | None  tastes: dict[str,
PositiveInt]  external_data = {  'id': 123,  'signup_ts': '2019-
06-01 12:22',  'tastes': {      'wine': 9,      b'cheese': 7,
      'cabbage': '1',  },}user = User(**external_data)
print(user.id)  #> 123print(user.model_dump())  """{
'id': 123,  'name': 'John Doe',  'signup_ts': datetime.datetime(2019,
6, 1, 12, 22),  'tastes': {'wine': 9, 'cheese': 7, 'cabbage': 1},}"""
```

If validation fails, Pydantic will raise an error with a breakdown of what was wrong:

Validation Error

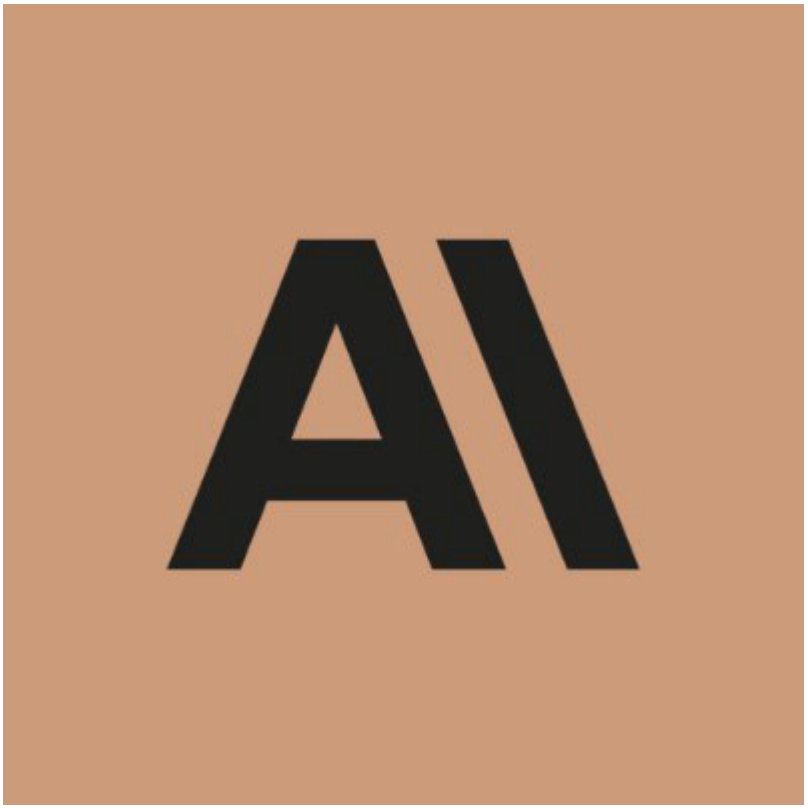
```
# continuing the above example...from datetime import datetimefrom
pydantic import BaseModel, PositiveInt, ValidationErrorclass
User(BaseModel):  id: int  name: str = 'John Doe'  signup_ts: datetime
| None  tastes: dict[str, PositiveInt]external_data = {'id': 'not an
int', 'tastes': {}}  try:  User(**external_data)  except
ValidationError as e:  print(e.errors())  """  [      {
'type': 'int_parsing',          'loc': ('id',),          'msg': 'Input
should be a valid integer, unable to parse string as an integer',
'input': 'not an int',          'url':
'https://errors.pydantic.dev/2/v/int_parsing',      },      {
'type': 'missing',          'loc': ('signup_ts',),          'msg':
'Field required',          'input': {'id': 'not an int', 'tastes': {}},


'url': 'https://errors.pydantic.dev/2/v/missing',      },  ]  """
```

Who is using Pydantic?

Hundreds of organisations and packages are using Pydantic. Some of the prominent companies and organizations around the world who are using Pydantic include:


amazon


A\


ASML


AstraZeneca


CISCO

DATADOG


HSBC


IBM


intel


INTUIT


ipcc
INTERGOVERNMENTAL PANEL ON
climate change

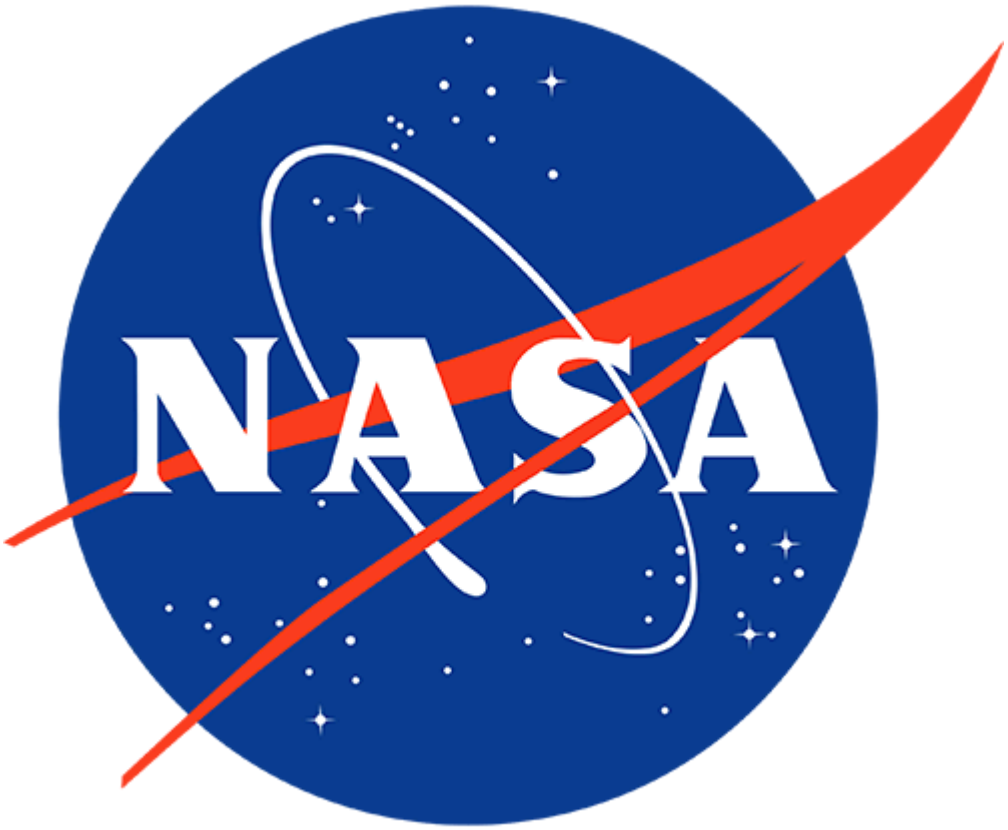

WMO
UNEP

JPMorgan
Chase & Co.


jupyter

Microsoft

MolSSI


NASA


NATIONAL SECURITY AGENCY
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NVIDIA

OpenAI

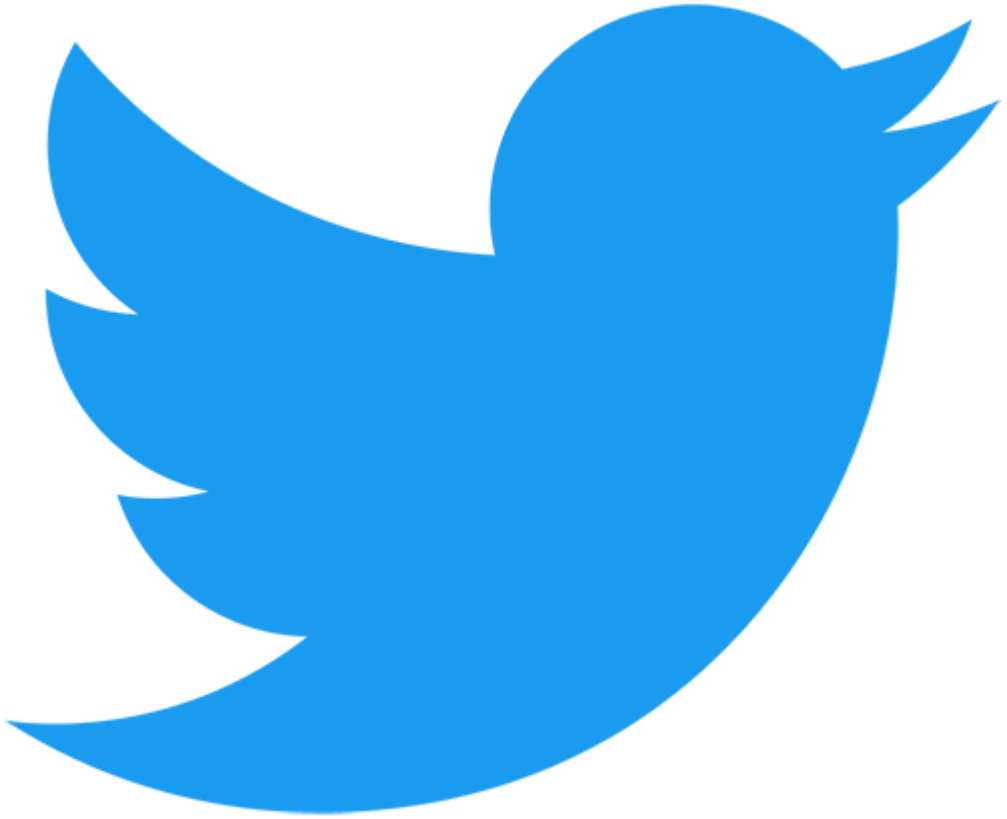

ORACLE


Qualcomm

Revolut


salesforce


twilio



For a more comprehensive list of open-source projects using Pydantic see the list of dependents on github, or you can find some awesome projects using Pydantic in awesome-pydantic.

Was this page helpful?